HR ANALYTICS DASHBOARD

SQL + POWER BI PROJECT

PROBLEM STATEMENT SUMMARY

- Organizations are struggling to get a handle on why employees leave how workers are spread out across the company and what makes employees happy on the job. This lack of understanding leads to strategies for keeping employees and bad decisions from HR.
- The HR data that is available has a lot of information on employees like where they come from what department they work in what they studied, what their job is and how satisfied they are with their work.. All this data is not being used to get useful insights that can help.
- The main issue here is employee attrition. Organizations are struggling with employee attrition. They are also struggling to understand workforce distribution.
- They are struggling to understand job satisfaction patterns. All these struggles lead to retention strategies. They also lead to HR decision-making.
- The available HR data contains information on employee demographics. It also contains information on departments.
- There is information on education. There is information on job roles. There is information, on satisfaction levels.
- It is not being effectively analyzed. It is not being used to generate insights.

DELIVERABLES

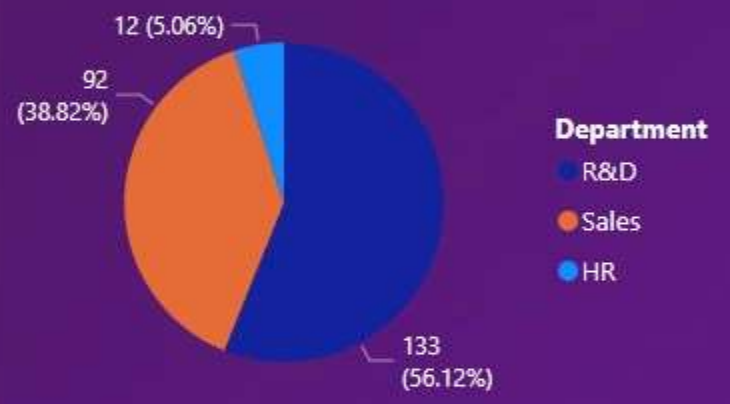
1. **Data Preparation & Modelling (MS-Excel)**: Clean and transform the raw dataset for analysis.
2. **Data Analysis (SQL)**: Organize the data into a structured format, simulate business transactions, and run queries to extract insights on customer segments, loyalty, and purchase drivers.
3. **Visualization & Insights (Power BI)**: Build an interactive dashboard that highlights key patterns and trends, enabling stakeholders to make data-driven decisions.
4. **Report and Presentation**: Write a clear project report summarizing your key findings and business recommendations. Prepare a presentation that visually communicates insights and actionable recommendations to stakeholders.
5. **GitHub Repository**: Include all Python scripts, SQL queries, and dashboard files in a well-structured repository.

HR ANALYTICS DASHBOARD

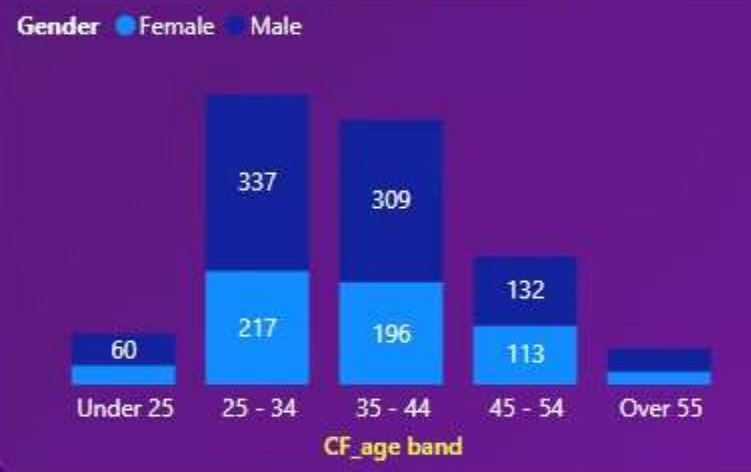
Associates Degree | Bachelor's Degree | Doctoral Degree | High School | Master's Degree

Overall Employees	Attrition	Attrition Rate	Active Employees	Average Age
1470	237	16.12%	1233	37

Department wise Attrition



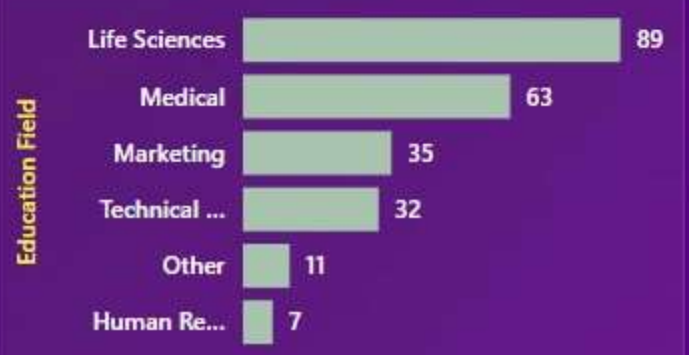
No. of Employees by Age group



Job Satisfaction Rating

Job Role	1	2	3	4	Total
Healthcare Representative	26	19	43	43	131
Human Resources	10	16	13	13	52
Laboratory Technician	56	48	75	80	259
Manager	21	21	27	33	102
Manufacturing Director	26	32	49	38	145
Research Director	15	16	27	22	80
Research Scientist	54	53	90	95	292
Sales Executive	69	54	91	112	326
Sales Representative	12	21	27	23	83

Attrition Count by Education Field



Attrition Rate by Gender for different Age groups

KEY BUSINESS INSIGHTS

- **Attrition Rate is 16.12% (237 employees)** → This is **not low**, it's a retention problem.
- **Sales (38.82%) and R&D (56.12%) dominate attrition** → Core business functions are bleeding talent.
- **Age group 25–34 has the highest employee count AND highest attrition (112)** → Mid-career instability.
- **Life Sciences (89) and Medical (63) show highest attrition by education** → Specialized talent loss.
- **Sales Executive (326 total) has high dissatisfaction and attrition exposure** → High-risk role.
- **Job satisfaction varies heavily across roles** → Not a company-wide issue, it's role-specific.
- **Male employees show slightly higher attrition in most age groups** → Possible workload or role imbalance.

ROOT CAUSE ANALYSIS

- High Role Pressure in Key Departments

  Attrition is heavily concentrated in Sales and R&D, which are critical revenue and innovation functions.

  This indicates high performance pressure, tight deadlines, and workload stress contributing to employee exits.

- Lack of Career Growth for Mid-Level Employees

  Employees in the 25–34 age group show the highest attrition.

  This suggests limited career progression opportunities, slow promotions, or lack of skill development, leading to frequent job switches.

ROOT CAUSE ANALYSIS

- Uneven Job Satisfaction Across Roles

  Job satisfaction levels vary significantly between roles.

  Roles such as Sales Executive and HR show relatively lower satisfaction, indicating inconsistent management practices, incentives, or work conditions.

- Loss of Specialized Talent

  Higher attrition is observed among employees from Life Sciences and Medical backgrounds.

  This points to high market demand and competitive offers, making it difficult to retain skilled professionals.

ROOT CAUSE ANALYSIS

- Absence of Targeted Retention Strategies

  Attrition is concentrated in specific departments, roles, and demographics rather than being evenly distributed.

  This indicates the organization is relying on generic HR policies instead of data-driven, targeted retention strategies.

Conclusion

- The company is facing significant attrition (16.12%), especially in Sales and R&D departments.
- Mid-career employees (25–34) are the most affected group, indicating gaps in career growth and engagement.
- Attrition is not random — it is driven by role pressure, job dissatisfaction, and lack of targeted HR strategies.
- Immediate, data-driven interventions are required to reduce attrition and stabilize workforce performance.